



case of two or more battery connections, the equaliser, which incorporates a cell balancing scheme, prevents individual cells from becoming over-stressed. It helps to equalise the batteries and increases their life by more than 200 per cent. It also assists in providing longer backup time and takes into account the ageing and operating conditions of the cells.

## BATTERY MONITORING

Intelligent battery monitoring, another technology built into UPS and inverter systems, helps to monitor battery voltage, actual charging/discharging current, battery temperature and ambient temperature, and records the number

of charge and discharge cycles the battery goes through. It has an intelligent algorithm which can measure the health of the battery.

The user can examine this data in order to predict potential failures and take preventative measures.

The system monitors the state of charge of each cell. Also, it monitors less critical and low-cost applications. Switching circuits control the charge applied to each individual cell in the chain during the charging process to equalise the charge on all the cells in the pack.

## INTELLIGENT LOAD SHARING

This technology enables the user to prioritise different electrical loads. These systems have multiple outputs for connecting to loads and each of

**Intelligent load sharing helps to segregate the loads in an intelligent way. Once the mains power goes off, the battery will automatically cut off the lower-priority load and divert current to the higher-priority load. It will ensure that the user's highest-priority load runs smoothly all the time**

these prioritised outputs can be independently controlled. For instance, in a hospital, the top-priority load may be the equipment in the operation theatre. Similarly, in a retail chain, the IT server may be the first priority, so that the billing operations continue without any interruption. The hospital and the retail chain may also have lower-priority loads. For example, lighting could be a lower-priority load in some areas of the hospital, which can be turned off for some time.

Intelligent load sharing helps to segregate the loads in an intelligent way. Once the mains power fails and the load switches to battery mode, the battery level de-escalates to a certain level. The battery will then automatically cut off the lower-priority load and divert current to the higher-priority load. It will ensure that the user's highest-priority load runs smoothly all the time. The optimisation of the load is based on priority levels. Such a system is the need of the hour in all critical application areas like hospitals, retail chain stores, shopping malls, etc.

## HIGH-FREQUENCY DEVICES

Today, the trend is to create products based on high frequency, which are sleek and smaller in size but are still EMI- and EMC-compatible. High-frequency-based UPS and inverters render numerous advantages like reduced overall system cost, less weight and improved system performance. However, in India, this system has been a failure so far because manufacturers have not designed these products well.

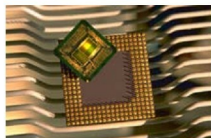
High-frequency products emit high-frequency radiation as power de-



vices are switched at high speed. They therefore need to be guarded against electromagnetic interference (EMI) and are designed for electromagnetic compatibility (EMC).

## CONFIGURABLE PROCESSOR

Usually, UPS and inverters lack a system where the hardware can be automatically configured. Now, manufacturers are going in for con-



figurable processors, through which many hardware blocks (such as pulse width modulators, analogue-to-digital conversion channels and general-purpose input/outputs) can be configured dynamically. ●

Information provided by Venkat Rajaraman, CEO, Su-Kam Power Systems